ClimateScope: Visualizing Global Weather Trends and Extreme Events

Milestone1 (Data Preparation & Initial Analysis)

1. Introduction

This document summarizes the data preparation, cleaning, and initial exploratory analysis conducted in Milestone 1 of the ClimateScope project. The primary objective of this phase was to acquire and preprocess global weather data, validate its quality, and ensure readiness for further statistical analysis and visualization. The dataset used in this project was obtained from the Global Weather Repository, available on Kaggle.

2. Dataset Description

- **Dataset Name:** Global Weather Repository
- **Source:** Kaggle
- **File Format:** CSV
- **Total Records: 125,501**
- **Total Features: 41**
- **Geographic Coverage:** Global
- **Temporal Reference:** Epoch-based timestamps (last_updated_epoch)

The dataset includes meteorological, air quality, and astronomical variables collected across diverse global locations.

3. Dataset Structure & Data Types

The dataset consists of mixed data types:

- **Float64:** 24 columns
- **Int64:** 7 columns
- **Object:** 10 columns

Table 1 Major Feature Groups

Category	Description
Location	Country, location name, latitude, longitude
Temporal	last_updated_epoch
Weather	Temperature, wind, pressure, humidity, precipitation
Air Quality	CO, NO_2, O_3, SO_2, PM2.5, PM10
Astronomical	Sunrise, sunset, moon phase

4. Data Quality Assessment

4.1 Missing Values

The dataset was assessed for missing values across all features. All meteorological, geographic, and air quality variables were found to be complete with no missing observations. The timezone column contained missing values across all records and was identified as non-essential for climate trend analysis. This column was therefore excluded from further processing. After this adjustment, the dataset contained no missing values in critical fields.

4.2 Duplicate Records

- Duplicate entries were checked and removed where identified
- This ensured accuracy during aggregation and trend analysis

4.3 Anomaly Validation

- Temperature, pressure, humidity, and wind values were validated against realistic meteorological ranges
- Only physically plausible observations were retained

5. Data Cleaning & Preprocessing

The following preprocessing steps were performed:

- Removal of duplicate records
- Conversion of the last_updated timestamp column into datetime format and usage as the primary temporal index for time-series analysis
- Removal of the last_updated string column to eliminate redundancy
- Conversion of epoch timestamps into datetime format
- Extraction of year and month features
- Standardization of units (Celsius, km/h, mm)
- Removal of redundant unit columns
- Renaming columns for clarity and consistency

Rationale:

Epoch-based timestamps provide a consistent, efficient, and unambiguous representation of time, making them ideal for time-series analysis.

6. Data Aggregation

To enable seasonal analysis and efficient visualization:

- Weather observations were aggregated to **monthly averages**
- Grouping was performed by **country, year, and month**

This transformation preserves climate patterns while reducing dataset complexity.

7. Season Identification

To support seasonal climate analysis, a new categorical feature called season was created by mapping calendar months to climatological seasons. Each monthly observation was assigned a season based on its corresponding month:

- Winter: December – February
- Spring: March – May
- Summer: June – August
- Autumn: September – November

This seasonal labeling enables comparative analysis of weather patterns across different periods of the year for individual countries and forms the basis for seasonal trend visualization.

8. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to validate data quality and understand basic variable distributions. The goal was not to derive deep insights but to confirm dataset reliability and readiness for further analysis.

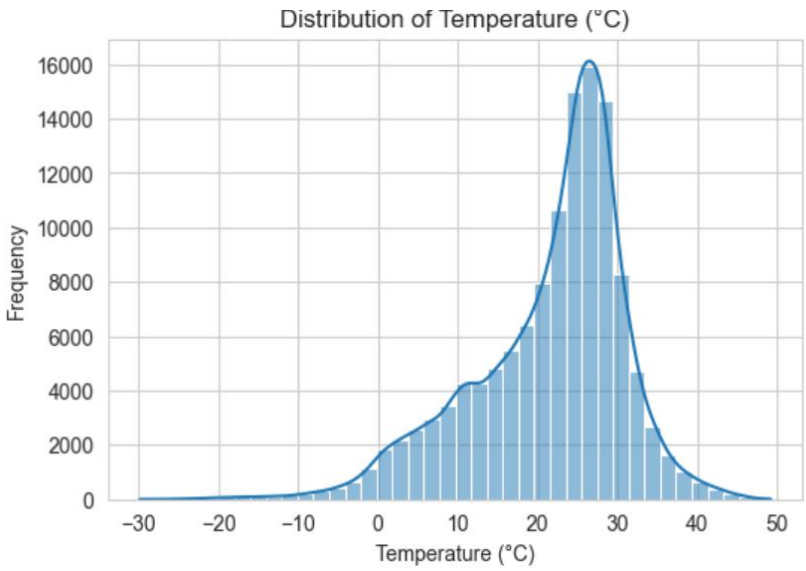

Fig 1: Distribution of Temperature (°C)

What this image shows:

- Temperature values fall within realistic climatic ranges
- No abnormal or corrupted values are present
- Data is suitable for climate trend analysis

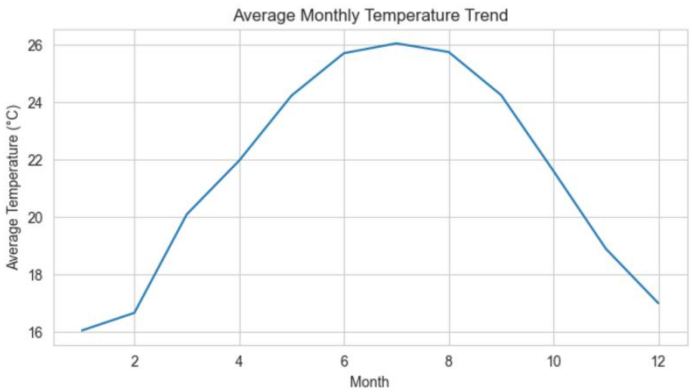

Fig 2: Average Monthly Temperature Trend

What this image shows:

- Clear seasonal temperature patterns
- Successful extraction of temporal features (year & month)
- Correct monthly aggregation logic

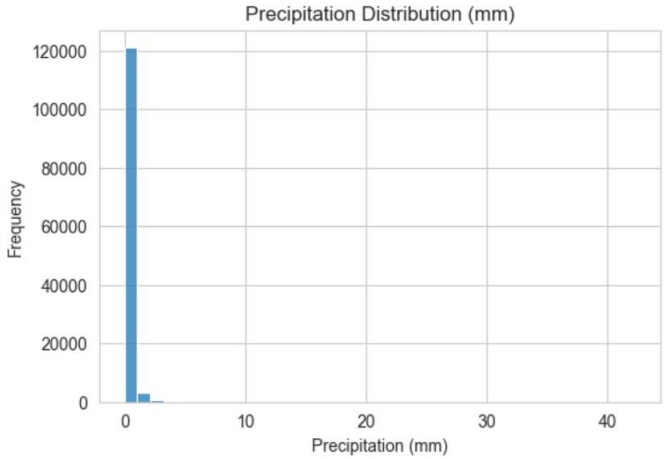

Fig 3: Precipitation Distribution (mm)

What this image shows:

- Precipitation data is right-skewed, as expected
- Presence of many zero-rainfall observations
- Confirms realism of rainfall measurements

Seasonal exploratory analysis was also performed using the derived season feature to validate seasonal temperature and precipitation behavior.

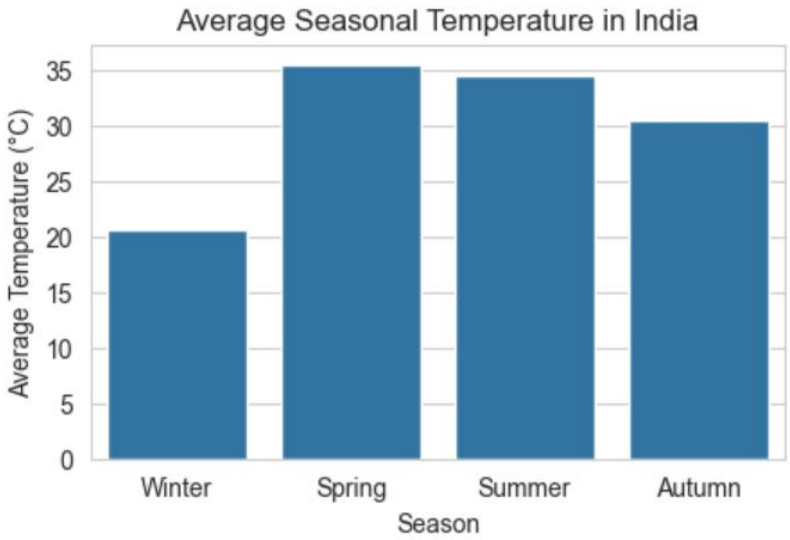

Fig 4: Average Seasonal Temperature by Season (Country-wise)

What this image shows:

- Clear variation in temperature across seasons
- Summer months exhibit higher average temperatures
- Confirms correctness of season classification

9. Final Processed Dataset

- File Name: weather_cleaned.csv
- Storage Location: data/processed/
- Granularity: Monthly aggregated data
- Format: CSV
- Additional Feature: season (Categorical variable representing climatological season)

Table 2 Final Dataset Schema

Column	Description
country	Country name
year	Observation year
month	Observation month
temperature_celsius	Mean monthly temperature (°C)
humidity	Mean monthly humidity (%)
precip_mm	Mean monthly precipitation (mm)
wind_kph	Mean monthly wind speed (kph)
uv_index	Mean monthly UV index

air_quality_PM2.5	Mean monthly PM2.5 concentration
Season	Climatological season (Winter, Spring, Summer, Autumn)

Conclusion

Milestone 1 was successfully completed. The Global Weather Repository dataset was thoroughly validated, cleaned, and transformed into an analysis-ready format. Seasonal labeling further enhances the dataset's suitability for climate trend analysis. Exploratory Data Analysis confirmed realistic distributions, complete data coverage, and correct temporal aggregation. The processed dataset provides a robust foundation for identifying seasonal trends, regional variations, and extreme weather events in subsequent milestones.

Next Steps

- Perform statistical and correlation analysis
- Identify seasonal patterns and extreme events
- Design interactive visualizations
- Develop the ClimateScope dashboard using Plotly and Streamlit